

# Xuan Zhou

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## EDUCATION

### University of California, San Diego

*M.S. in Computer Science (CSE)*

Expected Dec 2027

*La Jolla, CA*

### University of Minnesota, Twin Cities

*B.A. in Mathematics & Computer Science, College of Liberal Arts* **GPA: 3.83/4.0**

May 2026

*Minneapolis, MN*

- **Relevant Coursework:** Mathematics of Machine Learning & Data Analysis I & II (MATH 5465/5466), Machine Learning Fundamentals (CSCI 5521), Algorithms & Data Structures, Advanced Programming Principles, Operating Systems, Computer Security, Cryptology, Applied Linear Algebra, Intelligent MR Systems, Directed Research in CS.
- **Honors: Phi Beta Kappa** (Alpha of Minnesota Chapter, inducted 2026); Prof. Hans Dalaker Scholarship (2025); Dean's List (Fall 2024, Spring 2025).
- **Teaching:** Grader for **MATH 5465 & MATH 5466 – Mathematics of Machine Learning and Data Analysis I & II**, Sep 2025 – May 2026.

## TECHNICAL SKILLS

**Languages:** Python, C/C++, Go, Java, JavaScript/TypeScript, SQL, HTML/CSS, OCaml

**ML & AI:** PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, LangChain, LangGraph, Ollama, vLLM (LLM deployment & throughput tuning), Multi-Agent Systems, LLM Fine-tuning

**Frameworks & Tools:** NVIDIA Sionna, Spring / SpringBoot, MyBatis, Vue.js, Docker, Git, Linux, MATLAB

**Databases:** MySQL, Redis, MongoDB, SQLite

## RESEARCH EXPERIENCE

### UCSD — Embodied AI Project

*Latent Action Models (LAM) for Embodied Agents*

Jun 2026 – Present

*La Jolla, CA*

- Learning latent action representations from manipulation video **without action labels** ( $z = f(o_t, o_{t+k})$ ): three public corpora (Assembly101, EgoVerse, AgiBot World, 400k windows) with hand-pose and end-effector supervision; cross-domain benchmark on RoboDojo simulation.
- Independently ran 20+ experiments across three series testing the team's hypotheses — auxiliary supervision, frozen linear readout probes, causal intervention audits — under a pre-registered evaluation scorecard.

### UMN NetLab — Undergraduate Research Assistant

*Project A<sup>2</sup>RAN: Agentic AI for Wireless RAN Operation* *Advisor: Prof. Zhi-Li Zhang*

Sep 2025 – May 2026

*Minneapolis, MN*

- Co-designing **A<sup>2</sup>RAN**, an LLM-powered **multi-agent framework** enabling operators to manage wireless RANs via natural language; agents autonomously perceive, reason, and act on tasks such as power allocation, load balancing, and base station deployment.
- Built the **Channel Probing Agent** on **NVIDIA Sionna**, generating high-fidelity channel matrices from real-time 3D environments and exposing them as a shared memory layer for downstream task agents.
- Built a **predictive digital-twin** pipeline on top of the Sionna environment: the twin renders the current scene and forecasts it seconds ahead, enabling **signal optimization against the predicted future** rather than the present.
- Developed the **Power Allocation Agent** that wraps convex-optimization solvers behind a natural-language interface, so the LLM can directly reason about and modify physical-layer parameters to maximize average coverage signal strength under a total-power budget.
- Orchestrated multi-agent collaboration (Master Agent for task decomposition; Load Balancing & Base Station Deployment Planning Agents) with **LangChain / LangGraph** and a locally deployed LLM via **Ollama**, enabling closed-loop autonomous network operation.

### AI-Assisted Medical Diagnosis Group — Research Lead

*Computer Vision for Prostate MRI Region Recognition*

Sep 2023 – Dec 2023

- Led a team of 4 undergraduates to build a CV pipeline for prostate region detection on MRI; used **3D Slicer** for slice extraction and preprocessing.
- Fine-tuned **YOLOv5** with hyperparameter search, improving detection mAP and contributing to an AI-assisted clinical diagnosis prototype.

*Image Classification with ResNet-50 vs. Vision Transformer*

- Benchmarked **ResNet-50** and **Vision Transformer (ViT)** on a custom trash-classification dataset; quantified the impact of ImageNet pretraining on downstream accuracy.
- **Co-first-author** publication at **ICAICE 2022** (SPIE Proc. Vol. 12610) — see Publications.

PROJECTS

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**Account Risk Insight System** | *Go, Vue.js, Casbin, SQLite, paramiko, WinRM*

- Led full-stack design and delivery of a privileged-account risk management platform governing **6,000+ network devices, 248 servers, 420 VMs**; Go backend with RESTful APIs + Vue.js frontend, multi-protocol remote management (**SSH, WinRM, Telnet, SNMP**).
- Engineered a **password vault** with Chinese-standard **SM2/SM3** encryption, **RBAC** authorization (Casbin) and **OTP-based 2FA**; built risk-analysis modules including weak-password detection, privilege-escalation detection, and automated password rotation.
- Passed security validation (**NSFOCUS RSAS** scanning, Nmap probing, third-party code audit) with **zero high-risk findings**; awarded at the National College Student Information Security Competition (2023).

PUBLICATIONS

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J. Liu, J. Sun, and **X. Zhou** (*co-first authors*), “Comparison of ResNet-50 and Vision Transformer models for trash classification,” *Proc. SPIE 12610, Third International Conference on Artificial Intelligence and Computer Engineering (ICAICE 2022)*, 1261021 (April 2023). doi:10.1117/12.2671208

**Sub-reviewer**, IEEE MSN 2026 (22nd Int’l Conf. on Mobility, Sensing and Networking) — reviewed 2 conference submissions.

AWARDS & CERTIFICATIONS

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**National Honorable Mention**, National College Student Information Security Competition — Aug 2023

**Provincial First Prize**, “Linghang Cup” Network & Information Security Competition — Sep 2023

**Provincial Second Prize**, Blue Bridge Cup Algorithm Competition — May 2022